

VISHAL RAJ

Software Development Engineer (Backend-leaning) | Node.js | REST APIs | DSA

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EDUCATION

National Institute of Technology, Andhra Pradesh

2022 – 2026

Bachelor of Technology in Electronics & Communication Engineering

CGPA: 8.54/10

EXPERIENCE

South Bihar Power Distribution Company Limited (SBPDCL) | Software Engineering Intern

May'25 – Jun'25

- Developed a production-grade **Car Marketplace platform** using **Next.js** and **Supabase (PostgreSQL)**, supporting **100+ car listings**, wishlisting, **10+ filters**, car detail pages, and end-to-end **test drive booking** workflows. [LINK](#)
- Integrated **role-based authentication and access control** with **Clerk** for **2 user roles** (buyers, dealership admins), including secure session handling, booking state management, and dealership **working hours** configuration.
- Created **admin/dealership dashboards** to monitor **inventory** (available/sold/unavailable) and test drive states (pending/confirmed/completed), and **automated car listing** using the **Google Gemini API (flash-1.5)** to extract structured details from uploaded images with a **confidence score for validation**, reducing manual data entry.

Indian Institute of Technology, Bhubaneswar | Research Intern (Remote)

Dec'24 – Jan'25

- Collaborated on a Microsoft-supported research internship to build a personalized travel planning system using structured geospatial and user-preference datasets, improving itinerary relevance and recommendation accuracy.
- Engineered rule-based backend logic to generate optimized **3, 5, and 7-day itineraries**. [CERTIFICATE](#)

PROJECTS

Realtime ChatApp | React, Socket.IO, Node.js, MongoDB

Mar'25 – May'25

- Designed and developed a **real-time messaging system** supporting **1:1 and group chats** using **Socket.IO**, enabling low-latency bidirectional communication with reliable message delivery. [LINK](#)
- Secured authentication and session management** with **JWT**, including signup/login flows, password hashing, and **role-protected RESTful APIs and WebSocket endpoints**, with middleware-based token validation.
- Modeled **MongoDB schemas and APIs** for users, conversations, messages, and implemented **media sharing** (images/files), downloads, and a fully **responsive UI** across desktop and mobile devices.

BookStore App | MERN Stack, Tailwind CSS

Jan'25 – Feb'25

- Designed and built a **full-stack online bookstore** using **React, Node.js, Express, and MongoDB**, managing **250+ books across 15+ genres**, with RESTful APIs for catalog browsing, search, and inventory management. [LINK](#)
- Delivered a **responsive UI** with **React and Tailwind CSS** and enforced **JWT-based authentication** with secure password hashing and **user-level authorization**, including protected routes and access control, to support free and paid book access.

SKILLS

Backend/Databases: Node.js, Express.js, REST APIs, JWT, Socket.IO, MongoDB, PostgreSQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Languages: C++, Python, SQL, Java (Basic)

Tools/Core CS: Git, GitHub, Vercel, Data Structures & Algorithms, OS, DBMS, OOP

ACHIEVEMENTS / CERTIFICATIONS

Problem Solving & DSA Practice | LeetCode, GeeksforGeeks

- Solved **500+ Data Structures and Algorithms problems** across Graphs, Dynamic Programming, Trees, and Greedy Algorithms, building strong problem-solving and algorithmic skills through regular practice in **C++**.

Top 10 Finalist | Inspira 24-Hour Hackathon 2025

[LINK](#)

- Built an end-to-end **Fake News Detection system** with a **FastAPI** backend serving a **fine-tuned MobileBERT** model and a **Vite + JavaScript** frontend, including a **Chrome extension** for in-browser predictions.
- Implemented **REST APIs with CORS support** for real-time inference and integrated the **Google Gemini API** to generate factual rewrites for flagged content, enabling a complete detect-and-correct workflow.